

Indian Standard

(Reaffirmed 2014)

(Reaffirmed 2019)

CODE OF PRACTICE FOR PACKAGING OF HOUSEHOLD ELECTRICAL APPLIANCES

UDC 64.06.83:621.798



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INDIAN STANDARDS INSTITUTION

MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG

NEW DELHI 110002

Price Rs 5.00

July 1978

Indian Standard

CODE OF PRACTICE FOR PACKAGING OF HOUSEHOLD ELECTRICAL APPLIANCES

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CODE OF PRACTICE FOR PACKAGING OF HOUSEHOLD ELECTRICAL APPLIANCES

0. FOREWORD

0.1 This Indian Standard was adopted by the Indian Standards Institution on 30 January 1978, after the draft finalized by the Packaging of Engineering Goods Sectional Committee had been approved by the Marine, Cargo Movement and Packaging Division Council.

0.2 There is a considerable increase in the production and use of household electrical appliances in our country. The appliances have been designed and marketed for heating, cooling, cleaning, entertainment and as kitchen accessories.

0.3 Irrespective of the area of their use, these appliances are sensitive items, which require protection of their functional characteristics.

0.4 The package has to fulfil other objectives in addition to minimizing the damage to the contents and aesthetic consideration. It must communicate information like instructions for use, safety instructions and statutory information. It may have to fulfil another functional requirement of protecting the appliances after sale, when not in use.

0.5 Another eminent factor is the cost of the package. It is not desirable to pass unnecessary burden of cost to the ultimate consumer and therefore the materials which are available locally and commensurate with the functions expected to be performed have to be utilized.

1. SCOPE

1.1 This standard prescribes the practice for packaging of electrical appliances for household purposes, intended both for domestic distribution and for export.

1.1.1 This standard does not cover the following :

- a) Appliances designed for industrial purposes,
- b) Appliances made for medical treatment,
- c) Separate motors, and
- d) Portable tools.

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2. TERMINOLOGY

2.0 For the purpose of this standard, the following definition shall apply.

2.1 Household Appliances — An energy consuming device or equipment, other than a lamp, fixed or portable, in which electric energy is converted into light, heat, motion, or any other form of energy, or substantially changed in its electrical character and primarily used in households.

3. DETERMINATION OF PACKAGE

3.0 In the process of determining the appropriate package the following factors should be taken into consideration.

3.1 Product Characteristics — To start with, it is essential to know the details about the product to be packed. The physical characteristics of the product, such as size, mass, shape, materials of construction and strength, have to be known.

3.1.1 The prototype of a new product provides the basis for packaging considerations.

3.2 Conditions of Transport and Storage

3.2.1 Transport — The modes of transport normally employed, the hazards the product is likely to be subjected to during transportation and the distance to the ultimate destination affect the selection of a package.

3.2.2 Apart from the normal inherent hazards in transportation, handling is also responsible for transmitting damages to the packages. In export, the package is subjected to a much greater variety of handling methods and climatic conditions.

3.2.2.1 For all packs having a mass of contents more than 100 kg, provision has to be made to permit handling by power equipment. This may be by means of external skids, strips on the bottom of transport packages or by strips bolted to the bottom of the appliance and secured to inside face of the bottom of the package positioned in such a manner as to ensure adequate protection to the package and contents when handled with power equipment. If the strips or skids are secured to the inside face of the bottom of the transport packages it should be marked to indicate two opposite sides intended for fork lift truck handling.

3.2.3 Storage — The storage conditions likely to be encountered at various points, before it is delivered to the ultimate consumer, have to be considered.

3.2.3.1 In some cases the package may be called upon to extend protection even after reaching the ultimate destination and when the appliance is not in use.

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3.2.4 Stackability — Storage which may extend from short to long periods particularly when the storage space is small and not provided with modern amenities such as the stacking, offers hazards, which can lead to crushing of the lower packages. The other hazards are the dust and dirt which may spoil the communicative aspect of the package. The design of the storage bins and handling arrangements within the stores will depend upon the physical characteristics of the package and also the provision of handling equipment.

3.2.5 The package should be such that no deterioration of surface and mechanism occurs due to conditions prevailing in the surroundings. The package may be transported as :

- a) loose package;
- b) unitized (palletized) load; and
- c) containerized load, that is, the loose packages inside a freight container.

3.2.5.1 In order to achieve the simulation of actual distribution system in the design of package, tests conditions according to stipulations in Appendix A may be taken as a guideline.

3.3 Packaging Materials

3.3.1 Primary Protection — The most important function of packaging is the protection of the contents. This protection may be to the extent of wrapping in an acid-free tissue. Fine metal finishes shall be protected from scratches and sometimes from corrosion. In such cases the product is wrapped in tissue paper to prevent contact with the inner wall of the package, which might be abrasive. The vitreous enamel is another surface which, though, resistant to scratching is liable to damage if struck and needs protection against any such shock.

3.3.2 Outer Pack — Depending upon the product characteristics the outer pack may be of fibreboard or wood conforming to IS : 2771 (Part I)-1977*, IS : 2771 (Part II)-1975†, IS : 3071-1965‡, or IS : 1503-1967§, as the case may be. To select the type of the fibreboard box to be employed, IS : 6481-1971|| should be used for guidance.

3.3.3 Cushioning Material — The purpose of cushioning material is to reduce the intensity of shocks experienced by the outer container before they reach the contents. Two types of cushioning material are normally used, namely : (a) processed, and (b) unprocessed.

*Specification for fibreboard boxes: Part I Corrugated fibreboard boxes (first revision).

†Specification for fibreboard boxes: Part II Solid fibreboard boxes (first revision).

‡Specification for wooden crates.

§Specification for wooden packing cases (first revision).

||Guide for principal uses and styles of fibreboard containers.

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3.3.3.1 Processed material — Materials in this category may be rubber-based, plastic foams, paper-based or pneumatic. These materials ensure uniformity in characteristics; hence lend themselves to accurate assessment of dynamic characteristics.

3.3.3.2 Unprocessed material — In this category materials like wood wool, corrugated board, corrugated paper rolls, polystyrene strips, shredded paper may be included.

3.3.4 Recommended materials for the packaging of some of the appliances are given in Appendix B.

3.4 Disassembly — The manufacturer may disassemble the product for the purpose of packaging and shipment. The extent of disassembly may be decided by the following factors:

- a) Need for protection of fragile or projecting parts, and
- b) Reduction in cubage.

3.4.1 The parts disassembled should be adequately identified and instructions for proper assembly included in the package, preferably pictorial.

3.5 Economy — The selection of materials have to be matched by the economic considerations, the types of markets the product is expected to cater to, availability, their absolute cost and cost with respect to the packaged contents, and the functional utility of the package at the ultimate destination.

4. IDENTIFICATION

4.1 One of the essential functions of the package is to communicate. The communication may be of any, or all, of the following :

- a) Identification of the contents,
- b) Instructions for proper use, and
- c) Statutory information.

4.1.1 Contents — The communication may be pictorial or explanatory. It should be complete and convey an exact nature of the contents. Its conformity to accepted standards should be ensured.

4.1.2 Instructions for Use — For an efficient and safe use and maintenance of appliance, the manufacturer should incorporate instructions for use. These may be in one or more languages for easy compliance.

4.1.3 Statutory Information — Any information required by the statutory regulations shall be given on the package prominently.

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APPENDIX A

(Clause 3.2.5.1)

PACKAGE TESTS AND CONDITIONS THEREOF

A-1. DOMESTIC DISTRIBUTION OF CARGO

<i>Sl No.</i>	<i>Test</i>	<i>Ref to IS:</i>	<i>Hazards to be Simulated</i>	<i>Test Conditions</i>
(1)	(2)	(3)	(4)	(5)
i)	Drop test	7028 (Part IV)- 1973*	Drop by mis- handling	The drop height is decided depending on the bulk and the mass of the filled package (Drop height: 30 to 80 cm; number of drops: 7 or 8 times)
ii)	Vibration test	7028 (Part II)- 1973*	Vibration in transit	30 minutes each in horizontal and vertical positions

iii)	Revolving hexagonal drum test (2 m and 4 m)	—	Falls from loose cargo
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$$N = \frac{125 - 2.2 M}{\frac{L}{l} \times 1.25}$$

where

N = number of falls,
 M = mass of the filled package in kg,

L = maximum length of the filled package, and

l = minimum length of the filled package

*Performance tests for complete filled transport packages:

Part II Vibration test.

Part IV Vertical impact drop tests.

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<i>Sl No.</i>	<i>Test</i>	<i>Ref to IS:</i>	<i>Hazards to be Simulated</i>	<i>Test Conditions</i>
(1)	(2)	(3)	(4)	(5)
iv)	Compression test	7028 (Part VI)- 1973*	Pressure strength in transit and in warehouse	—
v)	Incline impact test	7028 (Part III)- 1973*	Collision of cargoes	Run-way distance: 100 to 150 cm
vi)	Spray test	7028 (Part VIII)- 1974*	Rain	Volume of water sprayed: 8 l/s

A-2. TESTS AND CONDITIONS OF EXPORT CARGOES

A-2.1 Export cargo tests are conducted in the same way as domestic cargo tests. The only difference shall lie in the condition of drop test.

A-2.2 CONDITION OF DROP TEST FOR EXPORT CARGO

<i>Sl No.</i>	<i>Kind of Cargo</i>	<i>Test Conditions</i>
(1)	(2)	(3)
i)	Loose package	Height of fall: a) Where mechanical handling exists : 60 to 120 cm b) Where mechanical handling does not exist : 80 to 140 cm
ii)	Unitized (palletized) package	Base drop : 30 cm Corner drop : 20 cm Number of falls : 3
iii)	Containerized package	Same tests and conditions as domestic cargo

*Performance tests for complete filled transport packages:
Part III Horizontal impact test, inclined plane test and pendulum test.
Part VI Compression test.
Part VIII Water spray test.

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APPENDIX B

(Clause 3.3.4)

PACKAGING METHODS FOR ELECTRICAL PRODUCTS

<i>Sl No.</i>	<i>Product Category</i>	<i>Product</i>	<i>Packaging Method</i>	<i>Ref to IS :</i>
(1)	(2)	(3)	(4)	(5)
i)	Small appliances	Tape recorder, transistor radio, electric stove, juicer, blender, toaster, and other products	Box: Corrugated fibreboard carton Cushioning materials : Polystyrene foam and corrugated fibreboard Primary protection: Polythene bag	2771 (Part I)-1975*
ii)	Medium-size appliances	Electric fan, stereo, air-conditioner	Box: Corrugated fibreboard carton Cushioning materials: Polystyrene foam and corrugated fibreboard Primary protection: Polythene bag	2771 (Part I)-1975*
iii)	Major appliances	Refrigerator, washing machine, television, stereo, phonogram	Box: Corrugated fibreboard carton (Joints are used to fix outer and inner boxes) or wooden crates Cushioning materials: Polystyrene foam and corrugated fibreboard Primary protection: Polythene bags	2771 (Part I)-1975* or 3071-1965†

*Specification for fibreboard boxes: Part I Corrugated fibreboard boxes (*first revision*).

†Specification for wooden crates.

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<i>Sl No.</i>	<i>Product Category</i>	<i>Product</i>	<i>Packaging Method</i>	<i>Ref to IS:</i>
(1)	(2)	(3)	(4)	(5)
iv)	Heavy products	Electric boiler, furnace for heating	Box: Wooden frame box or corrugated fibreboard carton with wooden bottom sheet Cushioning materials: Felt Primary protection: Polythene bags	1503-1967* or 2771 (Part I)- 1975†

*Specification for wooden packing cases (*first revision*).

†Specification for fibreboard boxes: Part I Corrugated fibreboard boxes (*first revision*).